Proof. Let $a = \frac{1}{n}(\varepsilon_1 + \cdots + \varepsilon_n)$. If not all ε_i are equal, then $|a| < 1$. Moreover, since any algebraic conjugate of a root of unity is also a root of unity, $|a'| \leq 1$ for any algebraic conjugate a' of a. But the product of all algebraic conjugates of a is an integer. Since it has absolute value < 1, it must equal zero. Therefore, $a = 0$. $\square$

Proof of Theorem 5.4.4. Let $\dim V = n$. Let $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ be the eigenvalues of $\rho_V(g)$. They are roots of unity, so $\chi_V(g)$ is an algebraic integer. Also, by Proposition 5.3.2, $\frac{1}{n}|C|\chi_V(g)$ is an algebraic integer. Since $\gcd(n, |C|) = 1$, there exist integers a, b such that $a|C| + bn = 1$. This implies that

$$\frac{a|C|\chi_V(g)}{n} + b\chi_V(g) = \frac{\chi_V(g)}{n} = \frac{1}{n}(\varepsilon_1 + \cdots + \varepsilon_n)$$

is an algebraic integer. Thus, by Lemma 5.4.5, we get that either $\varepsilon_1 = \cdots = \varepsilon_n$ or $\varepsilon_1 + \cdots + \varepsilon_n = \chi_V(g) = 0$. In the first case, since $\rho_V(g)$ is diagonalizable, it must be scalar. In the second case, $\chi_V(g) = 0$. The theorem is proved. $\square$

Theorem 5.4.6. *Let G be a finite group, and let C be a conjugacy class in G of size p^k where p is a prime and $k > 0$. Then G has a proper nontrivial normal subgroup (i.e., G is not simple).*

Proof. Choose an element $g \in C$. Since $g \neq e$, by orthogonality of columns of the character table,

$$\sum_{V \in \operatorname{Irr} G} \dim V \chi_V(g) = 0. \tag{5.4.1}$$

We can divide $\operatorname{Irr} G$ into three parts:

(1) the trivial representation,

(2) D, the set of irreducible representations whose dimension is divisible by p, and

(3) N, the set of nontrivial irreducible representations whose dimension is not divisible by p.

Lemma 5.4.7. *There exists $V \in N$ such that $\chi_V(g) \neq 0$.*